Statistical Thinking for Data Science

Week 8: Conditional probability and Bayes' rule

What changes when you learn something

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Week 8: Conditional probability and Bayes' rule

Learning Objectives

By the end of this week you should be able to:

- compute conditional probabilities, and read a conditioning statement correctly in both directions;
- test independence, and explain why independence and mutual exclusivity are opposites rather than cousins;
- use the law of total probability and a tree diagram to break a problem into cases;
- apply Bayes' rule, and compute the same answer with natural frequencies and with odds;
- explain why a test that is right 99% of the time can be wrong most of the times it fires, and what that means for screening decisions;
- identify the prosecutor's fallacy and the base rate fallacy in a written claim.

8.1 Conditioning is just restricting the sample space

The conditional probability of A given B is

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}, \qquad P(B) > 0.$$

Nothing mysterious happens: you throw away the outcomes where B did not occur and renormalise so the survivors sum to one. Every conditional probability answers the question "among the cases where B holds, what fraction are also A?"

Rearranged, it gives the multiplication rule $P(A \cap B) = P(B)P(A \mid B)$, which is how you build probabilities of sequences: the chance the first borehole is barren and the second is ore is the chance of the first, times the chance of the second *given* the first.

The two directions are different questions

$P(\text{positive} \mid \text{faulty})$ is a property of the instrument. $P(\text{faulty} \mid \text{positive})$ is what the maintenance crew needs. They are related by Bayes' rule and are usually nowhere near each other in size. Most bad reasoning about evidence is one of these quietly swapped for the other.

8.2 Independence

A and B are **independent** when learning one tells you nothing about the other:

$$P(A \mid B) = P(A), \quad \text{equivalently} \quad P(A \cap B) = P(A)P(B).$$

Use the product form to test independence, because it is symmetric and it survives zero probabilities gracefully.

Two warnings, both of which cost real money:

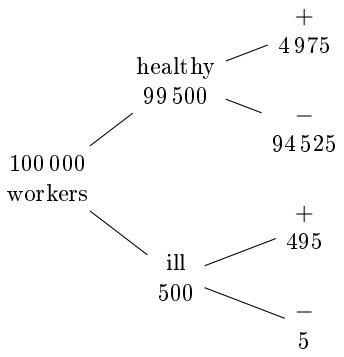
- **Mutually exclusive is the opposite of independent.** If A and B cannot both happen and both have positive probability, then knowing A occurred tells you B did not, which is the strongest possible dependence.
- **Independence is a claim about the world.** Two pumps are independent until they share a power line, a maintenance crew, a supplier or a flood. Reliability calculations fail in practice almost entirely because assumed independence was false, and the failure is always in the dangerous direction: simultaneous failures become much more likely than the model says.

8.3 Total probability: splitting into cases

If $B_1, \dots, B_k$ are mutually exclusive and cover everything, then for any A

$$P(A) = \sum_{i=1}^{k} P(B_i)\, P(A \mid B_i).$$

This is the formal version of a tree diagram: each branch is a case, you multiply along a branch and add across branches.



8.4 Bayes' rule

Combining the definition of conditioning with total probability:

$$P(B \mid A) = \frac{P(B)\, P(A \mid B)}{P(A)} = \frac{P(B)\, P(A \mid B)}{P(B)P(A \mid B) + P(B^c)P(A \mid B^c)}.$$

In words: the probability of a cause given the evidence is the probability of the cause, times how well the cause explains the evidence, divided by how likely the evidence was overall.

Example 1: the screening test, three ways

A condition affects 0.5% of the workforce. A screening test detects 99% of genuine cases and gives a false positive for 5% of healthy people. A worker tests positive. What is the probability he has the condition?

By formula

$$P(\text{ill} \mid +) = \frac{0.005 \times 0.99}{0.005 \times 0.99 + 0.995 \times 0.05} = \frac{0.00495}{0.05470} = 0.0905.$$

By natural frequencies

Think of 100 000 workers. 500 are ill, and 495 of them test positive. 99 500 are healthy, and 5% of those, 4 975, also test positive. So 5 470 positives in total, of which 495 are genuine: $495/5470 = 9.05\%$. This version is the one to use when explaining the result to anybody.

By odds

Prior odds of being ill are $1 : 199$. The likelihood ratio of a positive result is $0.99/0.05 = 19.8$. Multiply: posterior odds $= 19.8/199 = 0.0995$, so the probability is $0.0995/1.0995 = 0.0905$. The odds form shows the structure plainly: evidence multiplies the odds, it does not set them.

What to do about it

Retest. Among the 5 470 who tested positive the prevalence is now 9.05%, so a second independent test with the same characteristics gives $P = 0.0905 \times 0.99/(0.0905 \times 0.99 + 0.9095 \times 0.05) = 0.663$. That is why screening programmes are built in two stages: a cheap sensitive test first, a specific confirmatory test second.

Why 99% accurate is not 99% right

When the condition is rare, almost everybody who takes the test is healthy, so even a small false positive rate produces far more false alarms than true ones. The quality of a test cannot be judged without the prevalence in the group being tested. The same instrument is excellent in a high risk group and nearly useless as a mass screen.

8.4.1 The second test is rarely independent

The retest calculation above assumed the two tests fail independently. If both tests respond to the same underlying thing, such as a chemical that also appears in healthy people who smoke, a positive on the first makes a positive on the second more likely even in a healthy person, and the true posterior is lower than 0.663. Repeating the same test twice and claiming independence is one of the most common errors in practice. Use a confirmatory test of a different kind.

8.5 Two famous mistakes

8.5.1 The base rate fallacy

Ignoring $P(B)$ and judging only by how well the evidence fits the hypothesis. The screening example is the standard case: people hear "99% accurate" and answer 99%, which is wrong by a factor of eleven.

8.5.2 The prosecutor's fallacy

A match occurs with probability 1 in 10 000 by chance. The prosecutor says: the chance this person is innocent is 1 in 10 000. That is $P(\text{match} \mid \text{innocent})$ dressed up as $P(\text{innocent} \mid \text{match})$. In a city of a million people, about 100 innocent people also match. Without other evidence narrowing the pool, the match alone leaves a probability of guilt near 1%, not 99.99%.

Example 2: an exploration anomaly

A geochemical survey flags 2% of cells as anomalous. Of cells that genuinely contain mineralisation, 80% are flagged; of barren cells, 15% are flagged. A cell is flagged. Should you drill?

Solution

Taking 2% as the prior probability of mineralisation:

$$P(\text{ore} \mid \text{flag}) = \frac{0.02 \times 0.80}{0.02 \times 0.80 + 0.98 \times 0.15} = \frac{0.016}{0.163} = 0.098.$$

About one flagged cell in ten pays. Whether to drill is then an economics question, not a statistics one: drill if $0.098 \times$ (value of a discovery) exceeds the cost of a hole. The statistics gives the number; the decision needs the costs too.

8.6 Conditioning as the tool for everything that follows

Conditioning is not a topic, it is the mechanism behind most of the course:

- *Week 5*: a conditional distribution in a two-way table is exactly $P(A \mid B)$, and Simpson's paradox is what happens when you condition on the wrong thing or fail to condition at all.
- *Week 9*: a sampling distribution is the distribution of a statistic conditional on a sample size and a model.
- *Week 10*: a p-value is $P(\text{data this extreme} \mid \text{null hypothesis true})$, and reading it backwards is the base rate fallacy again.

- *Week 11*: a regression coefficient is a statement about the expected response conditional on the predictors.
- *Week 12*: Bayesian inference is this week's rule applied to parameters instead of to patients.

8.7 Doing it on a computer

These are a few lines of arithmetic, so the computer is useful mainly for checking intuition by simulation: generate a million artificial workers with the stated prevalence, simulate their test results, then count what fraction of the positives are genuinely ill. If the simulated fraction does not match your formula, one of the two is wrong, and it is worth knowing which before a report goes out.

8.8 Glossary

Conditional probability	$P(A \mid B)$, the probability of A among cases where B holds
Independence	$P(A \cap B) = P(A)P(B)$; learning one changes nothing about the other
Prior	the probability of a hypothesis before the evidence
Posterior	the probability after the evidence
Likelihood ratio	$P(\text{evidence} \mid H)/P(\text{evidence} \mid H^c)$, the factor by which evidence multiplies the odds
Sensitivity	$P(+ \mid \text{ill})$
Specificity	$P(- \mid \text{healthy})$
Positive predictive value	$P(\text{ill} \mid +)$, which depends on prevalence

8.9 Summary

Week 8 in six lines

1. Conditioning restricts the sample space and renormalises; that is all.
2. $P(A \mid B)$ and $P(B \mid A)$ are different questions with different answers.
3. Mutually exclusive and independent are opposites, not synonyms.
4. Bayes: posterior odds equal prior odds times the likelihood ratio.
5. A rare condition makes false positives outnumber true ones, however good the test.
6. Natural frequencies out of 100 000 persuade people that formulas do not.

Exercises

1. In a class, 60% take the data science elective, 30% take the mining elective and 12% take both. Find $P(\text{data science} \mid \text{mining})$ and $P(\text{mining} \mid \text{data science})$. Are the two electives independent?

2. A disease affects 1 in 200 workers. A test detects 99% of cases and falsely flags 5% of healthy people. Compute the probability of disease after a positive result, by formula and again with natural frequencies out of 100 000. Which version would you put in a letter to the workers, and why?

3. Continuing question 2, the worker takes a second, genuinely different test with the same characteristics and is positive again. Compute the new probability, then explain what would go wrong if the second test were simply the first one repeated.

4. Two pumps each fail with probability 0.1. Compute the probability that both fail, assuming independence. Then suppose a shared power failure, probability 0.03, knocks out both at once. Recompute, and say by what factor the independence assumption understated the risk.

5. A forensic match occurs by chance in 1 person in 50 000. The city has 2 million adults. A suspect matches. Write down the two conditional probabilities involved, compute roughly how many innocent people would match, and say what other evidence would change your answer.

6. A geochemical flag has sensitivity 0.8 and false positive rate 0.15, and 2% of cells are mineralised. Find the probability of ore given a flag. How high must the sensitivity go, with the false positive rate unchanged, to make the flag worth more than one hole in five?

7. A colleague writes: "95% of the machines that failed last year had skipped a service, so skipping a service gives you a 95% chance of failure." Write out both conditional probabilities and explain what extra number is needed.

8. Simulate the screening problem in question 2 with a million artificial workers, and report the simulated positive predictive value with its simulation error. Does it agree with the formula?

Answers to the odd-numbered exercises

1. $P(\text{DS} \mid \text{mining}) = 0.12/0.30 = 0.40$ and $P(\text{mining} \mid \text{DS}) = 0.12/0.60 = 0.20$. Independence would need $P(\text{both}) = 0.6 \times 0.3 = 0.18$, but it is 0.12, so the two choices are negatively associated: taking one makes the other less likely.

3. The prior is now 0.0905, so $P = 0.0905(0.99)/[0.0905(0.99) + 0.9095(0.05)] = 0.663$. Repeating the same test breaks the independence the calculation assumes: whatever made the first test misfire, a cross-reacting substance or a labelling error, is still present, so the second positive carries much less information and the true posterior is lower.

5. The two probabilities are $P(\text{match} \mid \text{innocent}) = 1/50\,000$ and $P(\text{innocent} \mid \text{match})$, which is what matters. About $2{,}000{,}000/50{,}000 = 40$ innocent people match, so the match alone leaves a probability of guilt near $1/41 \approx 2.4\%$. Evidence that narrows the pool before the match, such as being in the district that night, changes the prior and therefore the answer.

7. He has $P(\text{skipped} \mid \text{failed}) = 0.95$ and asserts $P(\text{failed} \mid \text{skipped})$. The missing number is the base rate: what fraction of all machines skipped a service. If most machines skip services and few fail, the claim is badly wrong.

Before next week

Work exercise 8, the simulation, and bring your code. Next week we stop asking about single events and start asking about estimates: how far a sample mean can be from the truth, and how to say so in public.